Monopolistic Competition

Answer key with worked solutions

At a profit-maximizing interior solution, MR = MC, L = (P - MC)/P = -1/ε, and MR = P(1 + 1/ε). Here ε is the signed own-price elasticity of the firm's demand. Qm and Qc denote monopoly and competitive output.

Source: Monopolistic Competition (1).pdf

#	Answer	Work / explanation
1	**c - 0.25**	At the profit-maximizing output, L = (P - MC)/P = -1/ε. With ε = -4, L = -1/(-4) = 0.25. Check: MR = MC = $9 and MR = P(1 + 1/ε) = 0.75P, so P = $12 and L = (12 - 9)/12 = 0.25.
2	**e - Bertrand**	In the standard Bertrand model with identical products and marginal costs, price competition drives P to MC. Therefore L = (P - MC)/P = 0, the lowest Lerner index among the listed cases.
3	**C - 0.60**	MR = 200 - 6Q and MC = 2Q. Set MR = MC: 200 - 6Q = 2Q gives Q = 25. Price is P = 200 - 3(25) = $125; MC = 2(25) = $50. L = (125 - 50)/125 = 0.60. Fixed cost does not change marginal cost.
4	**c, d**	The insurance brands illustrate monopolistic competition and product differentiation. Advertising, recognizable characters, and brand identity distinguish competing firms' products.
5	**b - $100**	At the profit-maximizing output, MC = MR = $50. The Lerner index is L = (P - MC)/P = -1/ε = -1/(-2) = 0.50. Thus (P - 50)/P = 0.50. Multiply by P: P - 50 = 0.50P, so 0.50P = 50 and P = $100.
6	**b - $400**	MR = 100 - 4Q; MC = 20. Monopoly: 100 - 4Q = 20 gives Qm = 20 and Pm = $60. Competition: 100 - 2Q = 20 gives Qc = 40. DWL = ½(40 - 20)(60 - 20) = $400.
7	**d - $140**	MR = 200 - 4Q; MC = 20 + 2Q. Set MR = MC: 200 - 4Q = 20 + 2Q, giving Qm = 30. Price is Pm = 200 - 2(30) = $140.
8	**d - $450**	From Question 7, Qm = 30 and Pm = $140. Competitive output solves 200 - 2Q = 20 + 2Q, giving Qc = 45. At Qm, MC = 20 + 2(30) = $80. DWL = ½(45 - 30)(140 - 80) = $450. Use the demand-MC gap at Qm as the triangle's height.
9	**a, b, c, d**	Regulatory capture can create entry barriers. Vertical integration can restrict rivals' access to inputs or distribution. Horizontal integration removes competitors. A price war can induce rivals to exit and increase future market power. Do not select e.
10	**c**	A firm in monopolistic competition with few substitutes faces less elastic demand and can sustain a larger price-cost margin. Since L = 1/|ε| at the optimum, a smaller elasticity magnitude means a higher Lerner index.
11	**a**	

#	Answer	Work / explanation
12	**a - 1/4**	At the profit-maximizing output, MC = MR = $6. Therefore L = (P - MC)/P = (8 - 6)/8 = 2/8 = 1/4.
13	**c**	Same as Question 10. Few substitutes make demand less elastic and allow a larger price-cost margin, giving the highest expected Lerner index among these choices.
14	**e - None** **P ≈ $133.33**	At the profit-maximizing output, MC = MR = $100. The Lerner index is L = (P - MC)/P = -1/ε = -1/(-4) = 0.25. Thus (P - 100)/P = 0.25. Multiply by P: P - 100 = 0.25P, so 0.75P = 100 and P = 100/0.75 ≈ $133.33. Select none of the above.
15	**A - 0.71**	MR = 300 - 10Q; MC = 2Q. Set MR = MC: 300 - 10Q = 2Q gives Q = 25. P = 300 - 5(25) = $175; MC = 2(25) = $50. L = (175 - 50)/175 = 125/175 ≈ 0.7143, or 0.71 to two decimal places.
16	**e - Monopolistic competition**	Advertising differentiates Lay's from competing brands and builds brand loyalty. This can make its demand less elastic and support a markup, illustrating nonprice competition in monopolistic competition.
17	**b - $100**	Same calculation as Question 5. MR = P(1 + 1/ε), so 50 = P(1 - 1/2) = 0.5P. Therefore P = $100.